Sai veginati

AI/ML Engineer

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SUMMARY

Machine Learning Engineer with **4+ years of experience** designing, deploying, and scaling **AI/ML solutions** across high-traffic production environments. Proven expertise in **MLOps, cloud-based ML deployment, NLP, recommendation systems, real-time data pipelines**, and **LLM-powered applications**. Skilled in **Python, TensorFlow, PySpark, Ray, Airflow, Kubernetes, Flink, Kafka, AWS,** and **IBM Cloud**. Adept at building **retrieval-augmented generation (RAG) systems, feature stores,** and **automated ML workflows**, delivering business impact and supporting **millions of users**. Strong track record of **collaborating with cross-functional teams** to deliver **scalable, production-ready ML solutions**.

TECHNICAL SKILLS

Programming & Scripting: Python (NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, PyTorch, Keras, NLTK, Matplotlib), R, Java, Scala, SQL (MySQL, PostgreSQL, Oracle, PL/SQL), NoSQL (MongoDB, Cassandra, DynamoDB)
Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning (CNN, RNN), Natural Language Processing (spaCy, GPT-4, Claude, LLaMA 2), Retrieval-Augmented Generation (RAG), Time Series Forecasting, Feature Engineering, Anomaly Detection, Reinforcement Learning, Generative AI, Ensemble Methods (XGBoost, LightGBM, Random Forest), Prompt Engineering, Recommendation Systems
MLOps & Model Deployment: Docker, Kubernetes, MLflow, Kubeflow, Jenkins, Git, CI/CD Pipelines, Flask, FastAPI, TorchServe, FBLearner, PyTorch Serve, Terraform, AWS Lambda, Serverless Architectures, Model Monitoring & Rollback
Big Data & Data Engineering: Apache Spark, Hadoop, Hive, Kafka, Apache Flink, Apache Airflow, Prefect, MapReduce, EMR, ETL/ELT Pipelines, Data Streaming, Data Pipeline Automation
Cloud Platforms & Data Warehousing: AWS (S3, Lambda, Redshift, EventBridge), IBM Cloud, Snowflake, Teradata, Watson Machine Learning
Data Analysis & Visualization: Exploratory Data Analysis (EDA), A/B Testing, Statistical Analysis, Data Wrangling, Tableau, Power BI, SSRS
Tools & Environments: Jupyter Notebook, RStudio, Linux Shell, RESTful APIs, Apache NiFi, ELK Stack, GitHub Actions, VS Code

WORK EXPERIENCE

Netflix | CA | ***Machine Learning Engineer*** **September 2024 – Present**

- Optimized **Retrieval-Augmented Generation** (**RAG**) systems, scaling to **10M+ QPS** with **FAISS vector stores** and **fine-tuned LLMs (GPT-4**, **Claude-2**), reducing recommendation errors by **22%** and improving content diversity by **10%**.
- Engineered a **real-time feature processing pipeline** with **Apache Flink** and **Kafka**, achieving **sub-500ms latency** on user interaction events (**clicks, skips, watch time**) and delivering instant personalization to **10M+ active users**.
- Established **end-to-end MLOps pipelines** using **MLflow** and **Kubeflow**, automating **model training**, validation, and deployment, which reduced **time-to-production** by **50%** and improved model rollback efficiency by **40%**.
- Designed and deployed a **serverless feature store** on **AWS (Lambda, DynamoDB, S3)** to manage and centralize real-time user embeddings, reducing feature duplication by **35%** and enhancing model freshness with **EventBridge-triggered updates**.
- Developed scalable **ML tooling** in **Python** (**NumPy, PySpark, Ray**) for automated **feature engineering** and **hyperparameter optimization**, cutting experiment iteration time by **40%** and increasing model accuracy through **systematic feature selection**.
- Implemented **NLP models** using **BERT** and **spaCy** to improve semantic matching and content comprehension for international content, resulting in a **15%** boost in **non-English title relevance**.
- Built a **multi-armed bandit testing framework** for recommendation systems, enabling adaptive **A/B testing** and achieving a **12%** improvement in user engagement metrics compared to **static allocation methods**.

IBM | India | ***Machine Learning Engineer*** **April 2020 – July 2023**

- Developed and deployed **supervised (XGBoost, LightGBM)** and **unsupervised (K-Means, DBSCAN)** models for **customer segmentation**, **fraud detection**, and **predictive maintenance**, improving anomaly detection accuracy by **30%** and reducing false positives by **22%**.
- Built and deployed **scalable ML models** on **IBM Cloud** using **Watson Studio**, **Watson Machine Learning**, and **Cloud Functions**, optimizing real-time inference for high-demand business applications.
- Automated end-to-end **ML workflows** using **Apache Airflow** by creating **custom DAGs** for data ingestion, feature engineering, and model retraining, reducing manual intervention by **60%**.
- Streamlined **CI/CD pipelines** for ML deployment using **Git**, **Docker**, **Kubernetes**, and **Jenkins**, cutting deployment time by **40%** and accelerating experimentation cycles.
- Engineered **machine learning solutions** in **Python** using **Scikit-learn**, **TensorFlow**, and **Keras** for **time-series forecasting** and **NLP tasks**, achieving **92% prediction accuracy** and reducing **training time** by **35%** through **optimized GPU utilization**.
- Collaborated with **cross-functional teams** to contribute to the **full SDLC** for **big data** and **ML systems**, integrating **Hadoop**, **Apache Spark**, and **IBM Cloud Pak** for Data to deliver scalable, production-ready **machine learning solutions**.

EDUCATION

Master's in Data Science | The University of Texas at Arlington, TX, USA.
Bachelor's in Information technology | Keshav Memorial Institute of Technology, Telangana, India.

PROJECT

Agentic AI System – Venue Image Generation (LangGraph, LLMs)

- Designed and built an agentic AI system that analyzes venue images, generates decorated outputs, self-evaluates results, and adapts behavior through retries and model fallbacks.
- Implemented a LangGraph-based workflow with shared state, conditional routing, and evaluation-driven decision-making rather than static prompt chaining.
- Used Claude Sonnet (Anthropic API) for vision-based analysis and LLM-driven self-evaluation within an agentic image generation workflow.
- Built a structured shared state schema to track inputs, intermediate reasoning artifacts, evaluation feedback, attempt history, and final outputs, enabling transparent and debuggable agent behavior.
- Developed prompt generation logic that dynamically incorporated venue analysis, color schemes, style intensity, and prior evaluation feedback to improve output quality across retries.
- Designed fallback mechanisms that switched generation strategies or models when quality scores dropped below thresholds, improving robustness under failure cases.
- Designed the system to be modular and extensible, allowing new tools, models, or evaluation criteria to be added without changing the core agent workflow.

CERTIFICATIONS

Generative AI Leader Certification — Google (2026)